

A solution to the long-standing problem of non-positive definite latent covariance matrices

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The latent R package

- Core functions written in C++
- Highly customizable models
- lavaan syntax for Structural Equation Models

```
Console Terminal x Background Jobs x
```

```
R 4.5.1 · ~/latent-master/ ↗
```

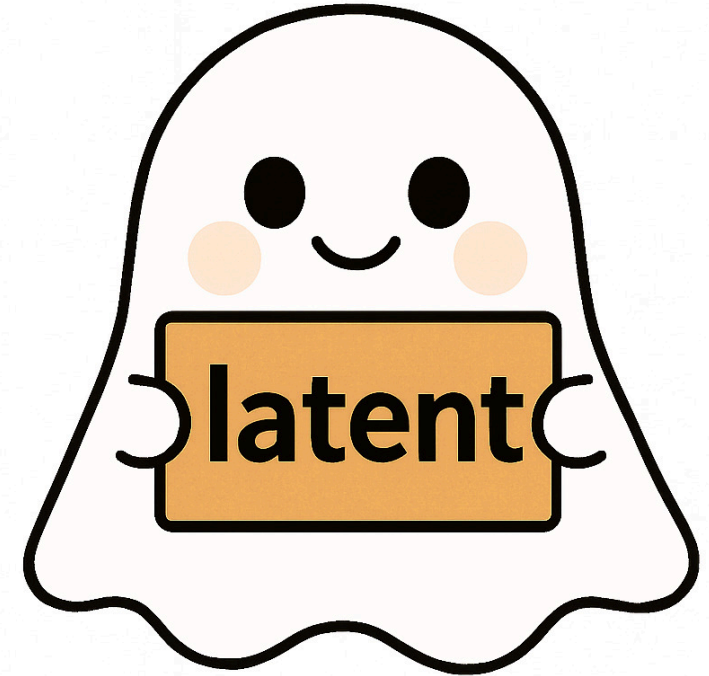
```
> library(latent)
```

```
👻 welcome to latent!
```

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version 0.1.0
```

```
Type 'citation("latent")' for citing this package in publications.
```

```
Report bugs at m.j.jimenezhenriquez@vu.nl  
For tutorials, visit marcosjnez.github.io/latent/
```



Example: Confirmatory Factor Analysis Example

Model fitting:

```
1 HS.model <- ' visual  =~ x1 + x2 + x3
2           textual =~ x4 + x5 + x6
3           speed   =~ x7 + x8 + x9 '
4 fit <- lcfa(model = HS.model, data = HolzingerSwineford1939)
```

Extract model information:

```
1 # Get fit indices:
2 lavaan::fitMeasures(fit)
3
4 # Inspect model objects:
5 lavaan::inspect(fit, what = "est", digits = 3) # Estimates
6 lavaan::inspect(fit, what = "se", digits = 3) # Standard error
```



Problem: Positive-definite constraints

In the factor model equation,

$$S = \Lambda \Psi \Lambda^\top + \Theta,$$

Latent correlations Ψ and covariances Θ should be positive-definite but...

Warning message:

```
lavaan->lav_object_post_check():  
  covariance matrix of latent variables is not  
  positive definite ; use lavInspect(fit, "cov.lv")  
  to investigate.
```

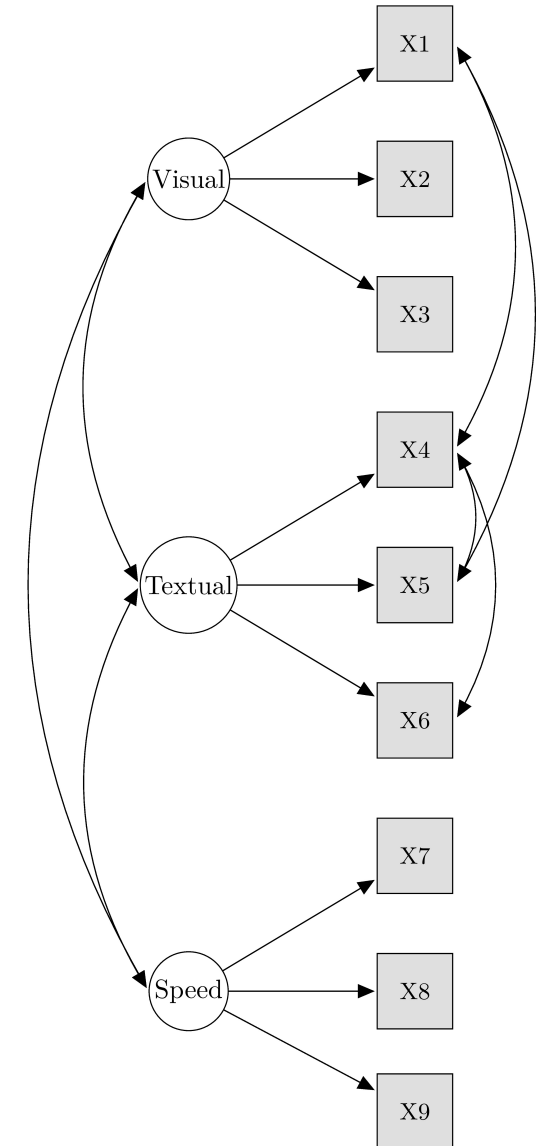
Positive-definite constraints (**lavaan** fails)

Let's force lavaan to fail for convergence:

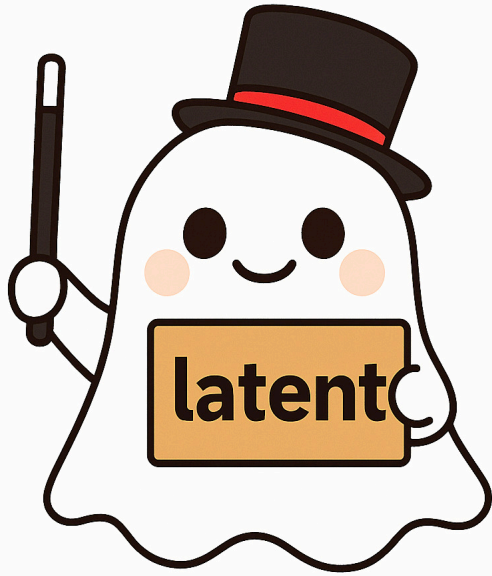
```
1 library(lavaan)
2 model <- 'visual =~ x1 + x2 + x3
3          textual =~ x4 + x5 + x6
4          speed  =~ x7 + x8 + x9
5          x1 =~ x5
6          x1 =~ x4
7          x4 =~ x5
8          x4 =~ x6'
9 fit <- cfa(data = HolzingerSwineford1939, model = model, estimator = "ml")
10 theta <- inspect(fit, what = "est")$theta
11 theta_determinant <- det(theta)
```

	x1	x2	x3	x4	x5	x6	x7	x8	x9
x1	0.62	0.00	0.00	0.11	0.09	0.00	0.00	0.00	0.00
x2	0.00	1.12	0.00	0.00	0.00	0.00	0.00	0.00	0.00
x3	0.00	0.00	0.79	0.00	0.00	0.00	0.00	0.00	0.00
x4	0.11	0.00	0.00	0.33	0.20	-0.26	0.00	0.00	0.00
x5	0.09	0.00	0.00	0.20	0.87	0.00	0.00	0.00	0.00
x6	0.00	0.00	0.00	-0.26	0.00	-0.10	0.00	0.00	0.00
x7	0.00	0.00	0.00	0.00	0.00	0.00	0.79	0.00	0.00
x8	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.47	0.00
x9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.59

det(theta) = -0.0096



Positive-definite constraints (**latent** converges)



```
1 fit <- lcfa(data = HolzingerSwineford1939, model = model,  
2           estimator = "ml", positive = TRUE)  
3 theta <- latInspect(fit, what = "est")$theta  
4 theta_determinant <- det(theta)
```

	x1	x2	x3	x4	x5	x6	x7	x8	x9
x1	0.61	-0.00	0.00	0.11	0.07	0.00	0.00	-0.00	-0.00
x2	-0.00	1.12	0.00	0.00	-0.00	-0.00	0.00	0.00	-0.00
x3	0.00	0.00	0.80	0.00	-0.00	0.00	0.00	-0.00	0.00
x4	0.11	0.00	0.00	0.43	0.19	-0.13	-0.00	0.00	0.00
x5	0.07	-0.00	-0.00	0.19	0.76	0.00	-0.00	0.00	0.00
x6	0.00	-0.00	0.00	-0.13	0.00	0.05	0.00	0.00	-0.00
x7	0.00	0.00	0.00	-0.00	-0.00	0.00	0.79	-0.00	0.00
x8	-0.00	0.00	-0.00	0.00	0.00	0.00	-0.00	0.47	0.00
x9	-0.00	-0.00	0.00	0.00	0.00	-0.00	0.00	0.00	0.58

det(theta) = 0.0002

Positive-semidefinite latent covariances



- Parameterize latent covariances as crossproducts:

$$\Psi = Y^\top Y$$
$$\Theta = U^\top U$$

- Constraint Y and U to be orthoblique:

$$X \in \mathbb{R}^{p \times p} : X^\top X = \text{sparse matrix}$$

$$\begin{bmatrix} 0.08 & 1.76 & 0.04 \\ -1.95 & -0.12 & -0.69 \\ -0.67 & -0.08 & 2.02 \end{bmatrix}^\top \begin{bmatrix} 0.08 & 1.76 & 0.04 \\ -1.95 & -0.12 & -0.69 \\ -0.67 & -0.08 & 2.02 \end{bmatrix} = \begin{bmatrix} 4.24 & 0.42 & 0.00 \\ 0.42 & 3.11 & 0.00 \\ 0.00 & 0.00 & 4.56 \end{bmatrix}$$

Positive-semidefinite latent correlations

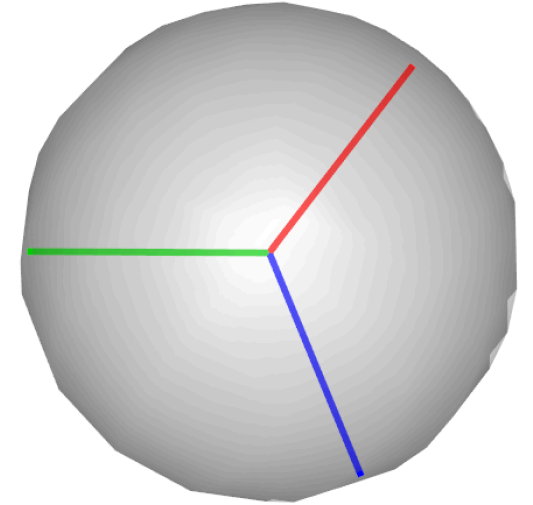


- Parameterize latent covariances as crossproducts:

$$\Psi = Y^{\top} Y$$
$$\Theta = U^{\top} U$$

- Constraint Y and U to be orthoblique:

$$X \in \mathbb{R}^{p \times p} : X^{\top} X = \text{sparse matrix}$$



$$\begin{bmatrix} 0.04 & 1.00 & 0.02 \\ -0.95 & -0.07 & -0.32 \\ -0.32 & -0.05 & 0.95 \end{bmatrix}^{\top} \begin{bmatrix} 0.04 & 1.00 & 0.02 \\ -0.95 & -0.07 & -0.32 \\ -0.32 & -0.05 & 0.95 \end{bmatrix} = \begin{bmatrix} 1.00 & 0.12 & 0.00 \\ 0.12 & 1.00 & 0.00 \\ 0.00 & 0.00 & 1.00 \end{bmatrix}$$

This is like factor rotation in EFA

$$S = \underbrace{\Lambda X^{-\top}}_{\Lambda^*} \underbrace{(X^\top X)}_{\Psi} \underbrace{(\Lambda X^{-\top})^\top}_{\Lambda^{*\top}} + \Theta$$
$$= \Lambda^* \Psi \Lambda^{*\top} + \Theta.$$

Type of constraint

- **Orthogonal rotation:**
 $X \in \mathbb{R}^{p \times p}$ such that $X^\top X = I$.
- **Oblique rotation:**
 $X \in \mathbb{R}^{p \times p}$ such that $X^\top X =$ dense matrix (with unit diagonal).
- **Orthoblique rotation:**
 $X \in \mathbb{R}^{p \times p}$ such that $X^\top X =$ sparse matrix (with unit diagonal).

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New factor rotation in EFA

PSYCHOMETRIKA—VOL. 66, NO. 2, 289–306
JUNE 2001

PSYCHOMETRIKA—VOL. 67, NO. 1, 7–20
MARCH 2002

A SIMPLE GENERAL PROCEDURE FOR ORTHOGONAL ROTATION

ROBERT I. JENNRICH

UNIVERSITY OF CALIFORNIA AT LOS ANGELES

A SIMPLE GENERAL METHOD FOR OBLIQUE ROTATION

ROBERT I. JENNRICH

UNIVERSITY OF CALIFORNIA AT LOS ANGELES

The orthoblique rotation for exploratory factor analysis

Marcos Jiménez

Mauricio Garnier-Villarreal

Vithor R. Franco

From positive-semidefinite to positive-definite

We may maximize the log likelihood subject to a weighted penalty

$$\operatorname{argmax}_{\theta} \ell(\theta) + w \cdot \text{penalty}$$

Suppose the penalty only involves Ψ .

- The parameterization $\Psi = Y^{\top} Y$ is an square, so $\det \Psi \geq 0$.
- Notice that $0 \leq \det(D^{-1/2} \Psi D^{-1/2}) \geq 1$, where $D = \text{diag}(\Psi)$.

Therefore, a good penalty term for Ψ becomes

$$\log \det(D^{-1/2} \Psi D^{-1/2}).$$



Summary

In the factor model $S = \Lambda \Psi \Lambda^\top + \Theta$,

1. the following parameterization will ensure, at least, positive-semidefinite matrices:

$$\Psi = Y^\top Y$$

$$\Theta = U^\top U,$$

where Y and U are **orthoblique** matrices (i.e., $X \in \mathbb{R}^{p \times p} : X^\top X = \text{sparse matrix}$).

2. Then, adding the **log det** of the standardized Ψ and Θ as a penalty to the log likelihood will encourage their positive-definiteness.

Cooking new stuff

- (Exploratory) Structural Equation Modeling
- Hidden Markov Models
- (Multidimensional) Item Response Theory

Release date? We don't know

Download the **beta version** at <https://github.com/Marcosjnez/latent>

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